

Passerines

PHYLUM	Chordata
CLASS	Aves
ORDER	Passeriformes
FAMILIES	131
SPECIES	6,430

Most of the world's bird species are passerines. They are sometimes known as perching birds because they have a unique type of foot that enables them to grip even the most slender branches. Passerines are also distinguished by the complex sounds, or songs, made by many species. These are produced using a vocal organ known as the syrinx (which is also found in other birds). Passerines are considered by many to be the most highly evolved of all birds, and many exhibit unusual intelligence. Most live in bushes and trees, but some are adapted to living on the ground; a few (such as swallows) lead an almost entirely aerial existence. Passerines are found in all terrestrial habitats, from arid desert to tropical rain forest. Many species are a familiar sight around buildings and in gardens, and new ones are being discovered all the time.

Anatomy

The passerines' specialized perching foot (see Perching) and well-developed voicebox (or syrinx, see Singing) are among their defining characteristics. These features apart, the members of this group are enormously varied. While a great number of them have subdued coloring, there are also many with spectacularly vivid and bizarre plumage—for example, the remarkable birds of paradise and the multicolored tanagers and finches. Males are often more brightly colored and patterned than females. Most passerines are small birds, but they range from the large, bulky crows and ravens, up to 25 in (65 cm) long, to the short-tailed pygmy tyrant, only 3 in (7 cm) long. Another highly variable feature of passerines is the shape of the bill, which often indicates a specific food preference (see below).



BILL SHAPE
The shape of a passerine's bill offers important clues to its diet. Species that feed on seeds (such as the greenfinch) often have a short, conical bill, while those that eat invertebrates (such as the wood warbler) tend to have a thin bill. Like many other predatory passerines, the great gray shrike has a hooked and notched bill for subduing relatively large prey. In sunbirds, which feed on nectar, the bill is long and thin, often with a downward curve, to help them reach inside flowers.

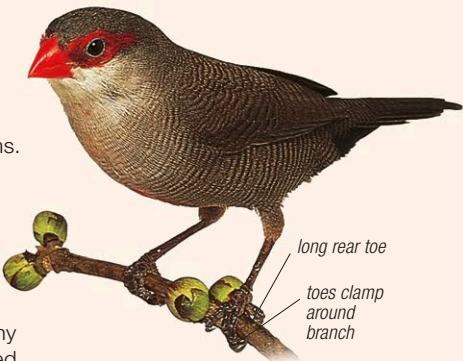
Singing

Passerines owe their singing ability to their complex syrinx, which is more highly developed than it is in other birds. Each species has its own distinctive song (or songs), a series of sounds uttered within a defined rhythm and structure. Many passerines have beautiful and complicated songs, among them the larks, wrens, thrushes, nightingales, and lyrebirds. In some species, the chicks learn to sing only by listening to adults of the same species; in others, the songs are innate and do not need to be learned. It is mainly the males that sing, either to claim and keep territory or to attract females.

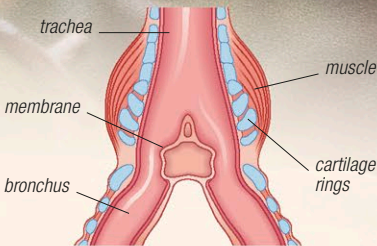


Perching

Passerines can perch securely on twigs, reeds, and even grass stems. Three of their 4 toes point forward and the other backward. The toes can move independently of each other, and the front toes can oppose the back one, which is especially strong. All 4 toes are level with each other, unlike in many nonpasserines, which have a raised hind toe. When a passerine lands on a perch, its weight causes tendons in the leg to tighten, clamping the toes tightly shut. This grip functions even when the bird is asleep. Despite these advantages, some passerines perch much less often than others. Larks, for example, are essentially terrestrial, and have relatively flat feet to give them balance when running.



PERCHING FEET
Like many passerines, the common waxbill has long, thin toes that are easily wrapped around a small stem. The backward-facing toe is strong and, in contrast to several nonpasserine groups, cannot be reversed. Most passerines have moderately curved, sharp claws that can grip many surfaces.



SINGING MALES
The male nightingale produces some of the most varied of all birdsongs, often delivering them with a characteristic crescendo. It sings both by day and by night, using its daytime songs mainly to mark its territory and deter rivals and its nocturnal songs to attract females.

SYRINX
The syrinx is located in the trachea. Membranes in the syrinx vibrate and produce sound as air passes over them. The bird uses muscles, which are attached to rings of cartilage, to modify the sound.